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AQA A-Level Physics — Required Practicals

There are 12 required practicals in the AQA A-Level Physics specification. These are assessed indirectly through exam questions that test your understanding of experimental methods, data analysis, sources of error, and practical skills. You should know each practical well enough to describe the method, draw and interpret graphs, identify errors, and suggest improvements.

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General Practical Skills for All RPs:

- **Variables:** Always identify the independent variable (what you change), dependent variable (what you measure), and control variables (what you keep constant).
- **Repeats:** Take at least 3 repeat measurements and calculate a mean to reduce random error.
- **Range:** Use at least 6 different values of the independent variable, spread across a wide range.

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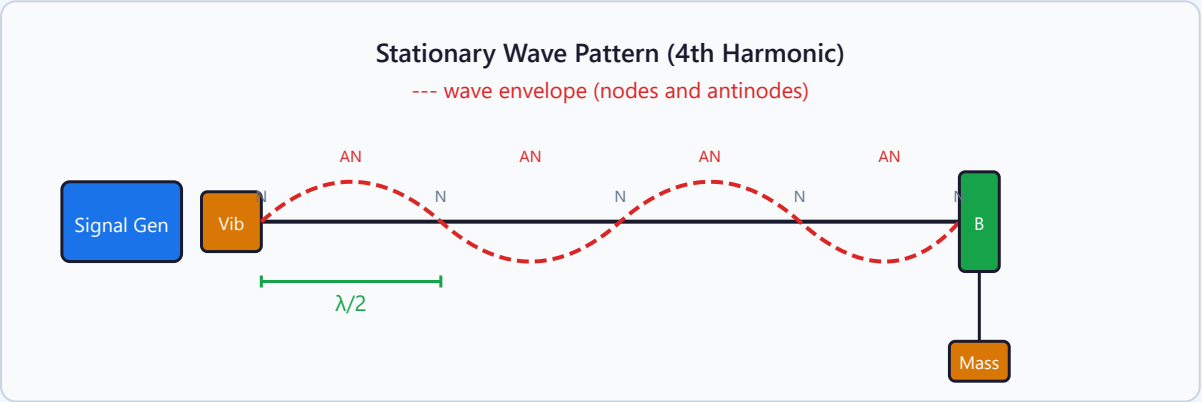
- **Graphs:** Plot the variable that gives a straight line. Use the gradient (and intercept if applicable) to determine the physical quantity. Draw a line of best fit and consider worst-case lines for uncertainty.
- **Uncertainties:** Calculate percentage uncertainties from instrument resolution or from the spread of repeats (half the range / mean × 100%).
- **Significant figures:** Give final answers to the same number of s.f. as the least precise measurement (usually 2 or 3 s.f.).

⚗ RP1: Stationary Waves on a String



Aim

To investigate the relationship between frequency, wavelength, tension, and mass per unit length for stationary waves on a string. Key equation: $f = (1/2L)\sqrt{(T/\mu)}$.

Apparatus

Signal generator, vibration generator (vibrator), string (nylon thread), pulley and bench clamp, slotted masses and hanger, metre ruler, wooden bridge (optional).

Method

1. Set up the vibrator at one end of the string. Pass the string over a pulley with a hanging mass to provide tension $T = mg$.

2. Turn on the signal generator and slowly increase the frequency until a clear stationary wave with one loop (fundamental mode, n = 1) forms.
3. Record the frequency f and measure the vibrating length L using a metre ruler. For the fundamental, $\lambda = 2L$.
4. Increase frequency to find higher harmonics (n = 2, 3, 4...). At the nth harmonic, there are n loops and $\lambda = 2L/n$.
5. To investigate the effect of tension: vary the hanging mass while keeping L constant. Record f at each tension.
6. To investigate the effect of length: vary L (using a bridge to change the vibrating length) at constant tension.

Safety

- Use a safety screen in case the string snaps under tension.
- Place a soft landing pad beneath the masses in case they fall.
- Ensure the vibrator is firmly clamped to prevent it moving.

Analysis

For the fundamental (n = 1): $f = (1/2L) \times \sqrt{(T/\mu)}$

- Plot **f vs 1/L** (keeping T and μ constant) — straight line through origin. Gradient = $(1/2)\sqrt{(T/\mu)}$.
- Plot **f vs √T** (keeping L and μ constant) — straight line through origin. Gradient = $1/(2L\sqrt{\mu})$.
- Plot **f^2 vs T** — straight line through origin. Gradient = $1/(4L^2\mu)$.

Worked Example: A string of length 0.80 m and mass per unit length 2.0×10^{-3} kg m^{-1} is under a tension of 10 N. Calculate the frequency of the fundamental mode.

Solution:

$f = (1/2L)\sqrt{(T/\mu)} = (1/(2 \times 0.80)) \times \sqrt{(10 / (2.0 \times 10^{-3}))}$
$= 0.625 \times \sqrt{(5000)} = 0.625 \times 70.7 =$ **44 Hz**

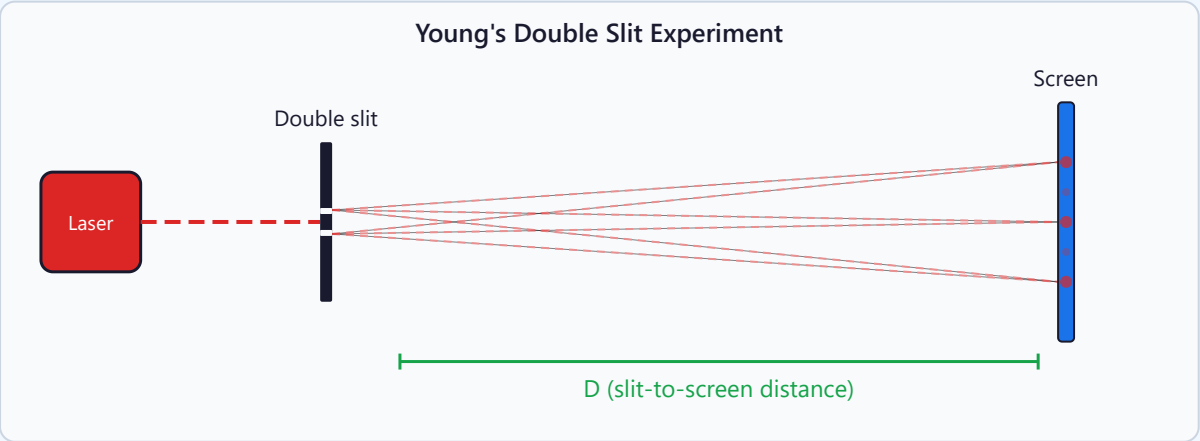
Sources of Error & Improvements

Source of Error	Type	Improvement

Difficult to judge exact resonance point	Random	Look for maximum amplitude; use a paper rider on the string (it falls off at antinodes at resonance)
String may not vibrate in a perfect plane	Random	Ensure vibrator is firmly clamped and aligned with the string
Ruler parallax when measuring L	Random	Use a ruler with mm divisions; read at eye level; measure from node to node
Mass of hanger not included in tension calculation	Systematic	Include the mass of the hanger in the total hanging mass: $T = (m_{masses} + m_{hanger})g$
String stretches under tension, changing μ	Systematic	Measure μ under the actual tension used; or use inextensible string

Exam Tip: Nodes are points of zero displacement; antinodes are points of maximum displacement. The fundamental (1st harmonic) has exactly one antinode and two nodes (at the fixed ends). The wavelength of the stationary wave equals 2L for the fundamental mode. At the nth harmonic: $f_n = nf_1$ and $\lambda_n = 2L/n$.

RP2: Young's Double Slit & Diffraction Grating



Aim

To determine the wavelength of light using (a) a double slit and (b) a diffraction grating.

Apparatus

Laser (or white light source with single slit and colour filter), double slit (known slit separation s), diffraction grating (known number of lines per mm), screen (white card), metre ruler, vernier callipers, tape measure for D.

Method (Double Slit)

1. Darken the room. Direct the laser beam through the double slit onto a distant screen (D > 1 m).
2. Observe the interference pattern of evenly spaced bright and dark fringes on the screen.
3. Measure the distance across several fringe spacings (e.g. across 10 bright fringes) and divide by the number of spacings to get the fringe spacing w. This reduces percentage uncertainty.
4. Measure the slit-to-screen distance D with a tape measure.
5. Calculate wavelength: $\boldsymbol{\lambda = ws/D}$, where s is the slit separation.

Method (Diffraction Grating)

1. Direct the laser through the diffraction grating onto a screen. Sharp, bright maxima appear at well-defined angles.
2. Measure the perpendicular distance x from the central maximum (n = 0) to the nth order maximum on the screen, and the distance D from the grating to the screen.
3. Calculate the angle: $\tan\theta = x/D$, hence $\theta = \arctan(x/D)$.
4. Calculate wavelength: $\boldsymbol{d \sin\theta = n\lambda}$, where d is the grating spacing (d = 1/N, where N is lines per metre).
5. Repeat for multiple orders (n = 1, 2, 3...) and average the calculated λ values.

Safety

- Never look directly into the laser beam — can cause permanent eye damage.
- Display a "Laser in use" warning sign on the door.
- Remove reflective objects (watches, jewellery, phones) from the beam path.
- Do not shine the laser at eye level of seated observers.

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Analysis

Double slit: $\lambda = ws/D$. The key uncertainty is usually in w (fringe spacing). Measuring across many fringes and dividing reduces this.

Diffraction grating: $d \sin\theta = n\lambda$. This is more accurate than the double slit because the maxima are sharper and more widely spaced, reducing the percentage uncertainty in θ.

Worked Example: A diffraction grating with 300 lines per mm is used with a laser. The 2nd order maximum is observed at 22.0 degrees from the central maximum. Calculate the wavelength of the laser light.

Solution:

$d = 1/(300 \times 10^3) = 1/(3.00 \times 10^5) = 3.33 \times 10^{-6}$ m

$d \sin\theta = n\lambda$

$\lambda = d \sin\theta / n = (3.33 \times 10^{-6} \times \sin 22.0°) / 2 = (3.33 \times 10^{-6} \times 0.3746) / 2 =$ **624 nm** (orange-red)

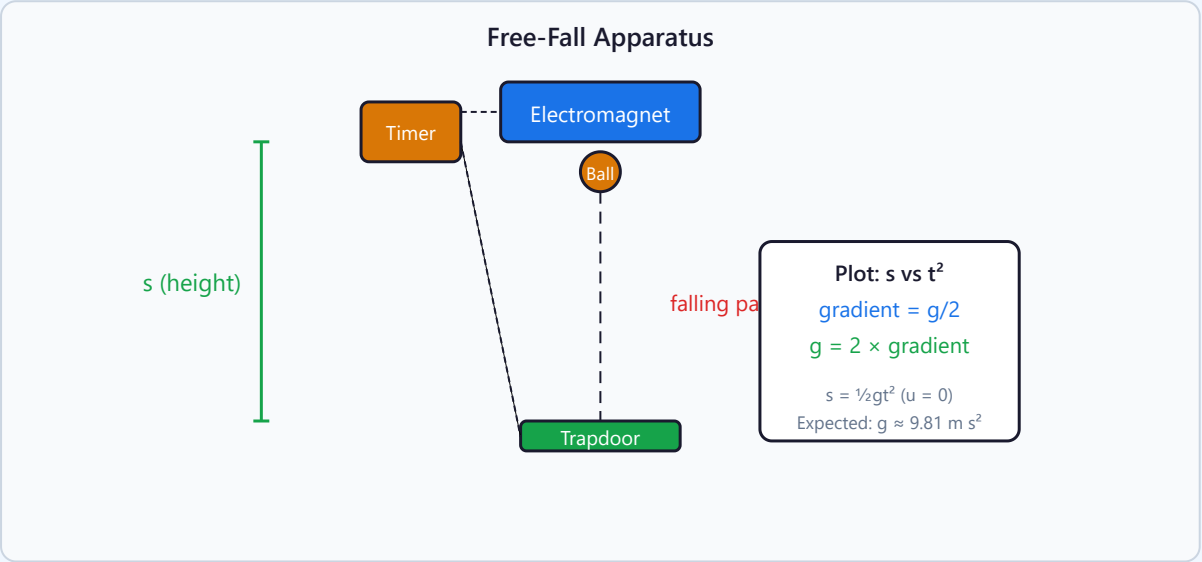
Sources of Error & Improvements

Source of Error	Type	Improvement
Fringes not sharp; hard to locate centres	Random	Use a diffraction grating for sharper maxima; darken the room fully
D measurement inaccurate	Random	Use D > 1 m to reduce percentage uncertainty; use a tape measure
Ruler not aligned with fringe pattern	Systematic	Ensure ruler is parallel to the fringe direction
Laser beam not perpendicular to slit/grating	Systematic	Align carefully; check that the zero-order maximum is directly ahead

Exam Tip: The diffraction grating is more precise than the double slit because the maxima are much sharper (due to constructive interference from hundreds of slits rather than just

two). Measuring across multiple fringes and dividing is essential for the double slit to reduce the percentage uncertainty in w. Always state the number of fringes measured.

RP3: Determination of g by Free-Fall



Aim

To determine the acceleration due to gravity g by timing the free fall of an object from rest through a known height.

Apparatus

Electromagnet connected to a switch and timer circuit, steel ball bearing, trapdoor switch connected to electronic timer, metre ruler (or 2 m ruler), clamp stand and boss.

Method

1. Clamp the electromagnet at a measured height s above the trapdoor.
2. Hold the steel ball on the electromagnet. When the switch is released, the timer starts automatically and the ball falls freely.
3. The ball hits the trapdoor, which opens and stops the timer. Record time t.

4. Repeat at least 3 times at each height and calculate the mean t.
5. Vary the height s (at least 6 different values from about 0.20 m to 1.00 m).
6. Plot s (y-axis) vs t^2 (x-axis). The gradient equals g/2, so g = 2 × gradient.

Safety

- Place a soft pad or cushion below the trapdoor to catch the ball and prevent bouncing.
- Ensure the clamp stand is stable (use a G-clamp at the base).
- Keep feet clear of the falling ball.

Analysis

Using $s = ut + \frac{1}{2}gt^2$ with u = 0 (released from rest): $s = \frac{1}{2}gt^2$.

Rearranging: $s = (g/2) \times t^2$. This is y = mx form, so plotting s vs t^2 gives a straight line through the origin with gradient = g/2.

Worked Example: A student obtains the following data and plots s vs t^2. The gradient of the best-fit line is 4.87 m s^{-2}. Calculate g and the percentage difference from the accepted value.

Solution:
g = 2 × gradient = 2 × 4.87 = **9.74 m s^{-2}**
% difference = |9.81 − 9.74| / 9.81 × 100% = **0.71%**
This is an excellent result. The small systematic error is likely due to residual magnetism causing a slight delay in release.

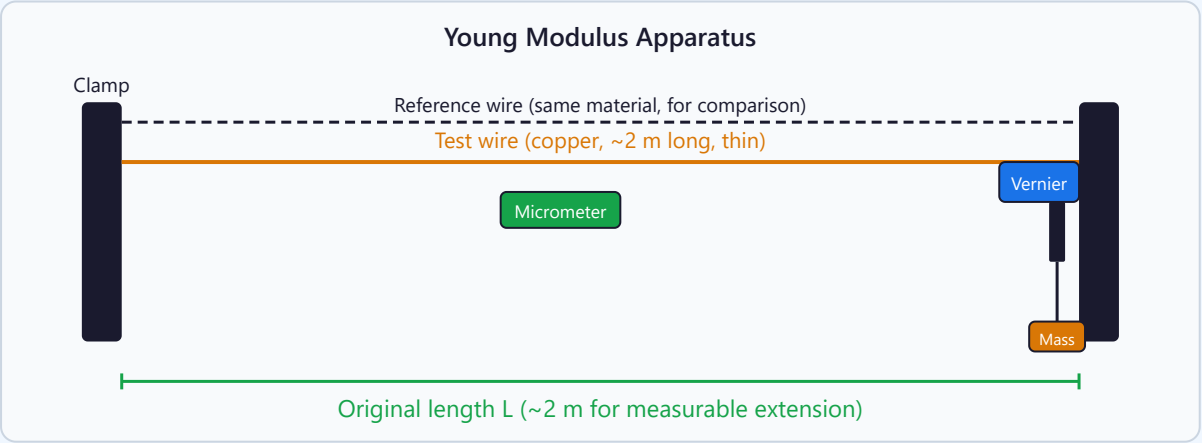
Sources of Error & Improvements

Source of Error	Type	Improvement
Ball may not release cleanly (residual magnetism holds it briefly)	Systematic (makes t too large, g too small)	Use a non-magnetic ball release mechanism; or account for delay by noting the graph's y-intercept

Height measurement uncertainty	Random	Use a long ruler; measure from the bottom of the ball to the trapdoor surface
Air resistance on the ball	Systematic (g appears smaller)	Use a dense, small, smooth ball to minimise air resistance
Timer starts late or stops early	Systematic	Use electronic timing with switch-triggered start and trapdoor-triggered stop

Exam Tip: If the s vs t^2 graph has a non-zero y-intercept, this indicates a systematic timing error (e.g. the ball is held slightly by residual magnetism before falling). The gradient is still valid for calculating g; the intercept represents the systematic offset. Using a graph with multiple data points averages out random errors and reveals systematic ones.

RP4: Young Modulus of a Wire



Aim

To determine the Young modulus E of a metal wire by measuring stress and strain. E = stress/strain = (F/A) / (ΔL/L).

Apparatus

Long thin wire (copper or steel, ~2 m), micrometer screw gauge, vernier scale or travelling microscope, slotted masses and hanger, metre ruler or tape measure, G-clamp and wooden blocks, safety goggles. A reference wire is useful to compensate for thermal expansion.

Method

1. Measure the wire diameter d with a micrometer at several points (at least 6 readings in different orientations). Calculate mean diameter, then area $A = \pi d^2/4$.
2. Measure the original length L of the wire (from clamp to marker) using a tape measure.
3. Add masses in steps (e.g. 100 g or 0.5 kg increments). After each addition, measure the extension ΔL using the vernier scale.
4. Record force F = mg and corresponding extension ΔL. Take at least 8 data points.
5. Continue up to a sensible maximum load (below the elastic limit).
6. Unload and check the wire returns to its original length (confirming elastic behaviour).

Safety

- Wear safety goggles — the wire may snap under high tension and whip back.
- Place a cushion or sand tray beneath the masses in case they fall.
- Do not exceed the elastic limit of the wire (watch for sudden increases in extension).

Analysis

Plot **stress (σ = F/A) vs strain (ε = ΔL/L)**. The gradient of the linear region = Young modulus E.

Alternatively, plot F vs ΔL. Gradient = EA/L, so E = (gradient × L)/A.

Typical values: $E_{copper} \approx 120$ GPa, $E_{steel} \approx 200$ GPa.

Worked Example: A copper wire of original length 2.50 m and diameter 0.28 mm has a 5.0 N load applied. The extension is measured as 1.8 mm. Calculate the Young modulus.

Solution:

$A = \pi(d/2)^2 = \pi \times (0.14 \times 10^{-3})^2 = 6.16 \times 10^{-8}\ m^2$

Stress = F/A = $5.0 / 6.16 \times 10^{-8} = 8.12 \times 10^{7}$ Pa

Strain = ΔL/L = $1.8 \times 10^{-3} / 2.50 = 7.20 \times 10^{-4}$

E = stress/strain = $8.12 \times 10^{7} / 7.20 \times 10^{-4}$ = **1.13×10^{11} Pa** (113 GPa, close to the accepted value of ~120 GPa)

Sources of Error & Improvements

Source of Error	Type	Improvement
Wire diameter varies along length	Random	Measure at 6+ points in different orientations; use the mean
Wire may not be straight initially (kinks)	Systematic	Apply a small initial load to straighten the wire before measuring
Extensions are very small and hard to measure	Random	Use a long, thin wire (maximises extension); use a vernier scale or travelling microscope
Thermal expansion of wire	Systematic	Use a reference wire alongside the test wire to compensate for temperature changes
Exceeding the elastic limit (permanent deformation)	Systematic	Unload after each reading to check the wire returns to original length

Exam Tip: The percentage uncertainty in diameter is doubled when calculating area (since $A = \pi d^2/4$). This is usually the largest source of uncertainty in the experiment. Use a long, thin wire to maximise the extension for a given stress, making measurements easier and reducing percentage uncertainty in ΔL.

⚗ RP5: Resistivity of a Wire

Aim: To determine the resistivity ρ of a wire (constantan) by measuring how resistance R varies with length L.

Key equation: $R = \rho L/A$. Plot R vs L: gradient = ρ/A, so ρ = gradient × A.

IV: Length L. **DV:** Resistance R (from V/I). **Controls:** Wire material, diameter, temperature.

Key apparatus: Constantan wire on metre ruler, ammeter (series), voltmeter (parallel), micrometer for diameter, low-voltage DC supply, switch.

Key points: Switch off between readings to prevent heating. Measure diameter at 6+ points. Biggest uncertainty is in d (doubled for area). Constantan is chosen because its resistivity barely changes with temperature. Expected $\rho \approx 4.9 \times 10^{-7}$ Ω m.

Full details including diagrams are in the 3.5 Electricity notes.

⚗ RP6: EMF and Internal Resistance

Aim: To determine the EMF ε and internal resistance r of a cell by plotting terminal PD (V) against current (I).

Key equation: $V = \varepsilon - Ir$. Plot V vs I: y-intercept = ε, gradient = −r.

IV: External resistance (varied by rheostat). **DV:** V and I. **Controls:** Same cell, temperature.

Key apparatus: Cell, variable resistor (rheostat), ammeter (series), voltmeter (across cell terminals), switch.

Key points: Open switch between readings to prevent cell running down. Do not short-circuit the cell. Typical: ε ≈ 1.5 V, r ≈ 0.5–1.5 Ω for an AA cell.

Full details including diagrams are in the 3.5 Electricity notes.

⚗ RP7: SHM — Mass-Spring and Pendulum

Aim: To investigate the factors affecting the period of oscillation and to verify the SHM equations.

Key equations: Mass-spring: $T = 2\pi\sqrt{(m/k)}$, so $T^2 = (4\pi^2/k)m$. Pendulum: $T = 2\pi\sqrt{(L/g)}$, so $T^2 = (4\pi^2/g)L$.

Mass-spring: Plot T^2 vs m. Gradient = $4\pi^2/k$. Find k.

Pendulum: Plot T^2 vs L. Gradient = $4\pi^2/g$. Find g.

Key points: Time 10+ oscillations and divide. Start counting from zero at the fiducial marker. Keep amplitudes small (< 10° for pendulum). Use a small, dense bob. Measure L to the centre of the bob.

Full details including diagrams are in the 3.6 Further Mechanics notes.

⚗ RP8: Boyle's Law and Charles's Law

Aim: To verify the gas laws: (a) Boyle's Law ($P \propto 1/V$ at constant T); (b) Charles's Law ($V \propto T$ at constant P).

Boyle's Law: Plot P vs 1/V — straight line through origin. Use a sealed gas syringe with pressure gauge.

Charles's Law: Plot V (or column length L) vs T (in kelvin) — straight line; extrapolate to find absolute zero. Use a capillary tube in a water bath.

Key points: Compress slowly for Boyle's (allow thermal equilibrium). Stir water bath for Charles's. Always convert T to kelvin. Friction in syringe is a common source of error.

Full details including diagrams are in the 3.6 Further Mechanics and Thermal Physics notes.

⚗ RP9: Capacitor Charge and Discharge

Aim: To investigate the charge and discharge of a capacitor through a resistor and determine the time constant $\tau = RC$.

Key equation: $V = V_0 e^{-t/RC}$. Plot ln(V) vs t: gradient = $-1/RC$.

Key apparatus: Capacitor (e.g. 1000 μF), resistor (e.g. 10–100 kΩ), DC supply, two-way switch, voltmeter or datalogger.

Key points: Choose R and C so τ is 10–60 s. Correct polarity for electrolytic capacitors. Use a datalogger for best results. After 5τ, the capacitor is effectively fully discharged.

Full details including diagrams are in the 3.7 Fields notes.

⚗ RP10: Force on a Current-Carrying Wire

Aim: To investigate F = BIL by measuring the force on a current-carrying wire in a magnetic field.

Key equation: F = BIL. Plot F vs I: gradient = BL, so B = gradient/L.

Key apparatus: Horseshoe magnet on a top-pan balance, stiff copper wire cradle clamped above, ammeter, variable DC supply.

Key points: The balance detects the reaction force (Newton's third law). Tare the balance before switching on current. Wire must be perpendicular to B for maximum force. F = Δm × g.

Full details including diagrams are in the 3.7 Fields notes.

⚗ RP11: AC Supply with Oscilloscope

Aim: To use an oscilloscope to measure the peak voltage V_0, frequency f, and period T of an AC supply.

Key equations: V_0 = divisions × V/div. T = divisions × time/div. f = 1/T. $V_{rms} = V_0/\sqrt{2}$.

Key apparatus: Oscilloscope, signal generator (or low-voltage AC supply), connecting leads.

Key points: Adjust Y-gain and time base for a clear display. V_{rms} is the DC equivalent that delivers the same power. UK mains: V_{rms} = 230 V, f = 50 Hz.

Full details including diagrams are in the 3.7 Fields notes.

⚗ RP12: Inverse-Square Law for Gamma Radiation

Aim: To verify the inverse-square law ($I \propto 1/d^2$) for gamma radiation from a point source.

Key equation: Corrected count rate $C \propto 1/d^2$. Plot C vs $1/d^2$: straight line through origin.

Key apparatus: Sealed gamma source (e.g. Co-60), GM tube and counter, metre ruler, stopwatch.

Key points: ALWAYS measure and subtract background count rate first. Use long counting times (≥ 60 s) because radioactive decay is random. Handle the source with tongs, never bare hands. Maximise distance, minimise exposure time. If the graph doesn't pass through the origin, there is a systematic error in d (correct with d_0).

Full details including diagrams are in the 3.8 Nuclear Physics notes.

Summary: All 12 Required Practicals at a Glance

RP	Topic	What to Plot	What Gradient Gives
1	Stationary waves	f vs 1/L or f vs √T	(1/2)√(T/μ) or 1/(2L√μ)
2	Double slit / grating	Measure w, D → λ = ws/D; or d sinθ = nλ	— (direct calculation)
3	Free-fall (g)	s vs t^2	g/2
4	Young modulus	Stress vs strain or F vs ΔL	E or EA/L
5	Resistivity	R vs L	ρ/A
6	EMF & int. resistance	V vs I	−r (y-intercept = ε)
7	SHM (spring/pendulum)	T^2 vs m or T^2 vs L	$4\pi^2/k$ or $4\pi^2/g$
8	Gas laws	P vs 1/V or V vs T(K)	nRT or nR/P
9	Capacitor discharge	ln(V) vs t	−1/RC
10	Force on wire	F vs I	BL
11	AC with oscilloscope	Read V_0 and T from screen	— (direct reading)
12	Inverse-square law	C vs $1/d^2$	Proportionality constant

Final Exam Tip: In practical questions, marks are given for: (1) correct identification of variables, (2) description of a clear method, (3) specific mention of apparatus and how measurements are taken, (4) stating what to plot and how to analyse the graph, (5) identifying and explaining sources of error, (6) suggesting specific improvements (not just "be more careful"). Always think about what limits the precision of your measurement and how you could reduce that uncertainty.